

STATISTICS • PART 2 • PROBLEM SET

# Worked Problems & Computations

*Formula, substitution and result shown separately for every step.*

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Companion to the Confidence Intervals & Hypothesis Testing deck

$$\bar{x} \pm t \cdot s / \sqrt{n}$$

# How every step on these slides is laid out

GIVEN

Sample size  
 **$n = 30$**

Sample mean  
 **$\bar{x} = 100,200$**

Population  $\sigma$   
**15,000**

1

**What is being found**

Formula

the rule written in symbols, before any numbers appear

Substitute

the same rule with the actual values put in place

Result

**the number this step produces, carried into the next one**

## Three lines, always

- The symbolic formula comes first, so the rule is learned before the arithmetic
- The substitution shows exactly which number went where
- The bold result is what the next step picks up

## A note on rounding

Intermediate values here are kept to four decimals and rounded only at the end. Answer keys often round the standard error first, so the last digit can differ by one. Both are correct.

## SECTION A

# Confidence Intervals

- One population, variance known and unknown
- Dependent samples
- Independent samples, both variance cases

# A 95% interval for data scientist salary

A sample of 30 data scientist salaries has a mean of \$100,200. Past data puts the population standard deviation at \$15,000, and salaries are normally distributed. Build a 95% confidence interval for the population mean.

|       |    |           |          |            |
|-------|----|-----------|----------|------------|
| GIVEN | n  | $\bar{x}$ | $\sigma$ | Confidence |
|       | 30 | \$100,200 | \$15,000 | 95%        |

|   |                |            |                                             |
|---|----------------|------------|---------------------------------------------|
| 1 | Standard error | Formula    | $SE = \sigma / \sqrt{n}$                    |
|   |                | Substitute | $SE = 15,000 / \sqrt{30} = 15,000 / 5.4772$ |
|   |                | Result     | <b>SE = 2,738.61</b>                        |

|   |                    |            |                                                                                    |
|---|--------------------|------------|------------------------------------------------------------------------------------|
| 2 | Reliability factor | Formula    | confidence = $1 - \alpha$ , then look up z at $\alpha/2$                           |
|   |                    | Substitute | $\alpha = 0.05$ , $\alpha/2 = 0.025$ , table entry 0.9750 $\rightarrow 1.9 + 0.06$ |
|   |                    | Result     | <b>z = 1.96</b>                                                                    |

# Margin of error and the two limits

## 3 Margin of error

|            |                                   |
|------------|-----------------------------------|
| Formula    | $ME = z \times SE$                |
| Substitute | $ME = 1.96 \times 2,738.61$       |
| Result     | <b><math>ME = 5,367.68</math></b> |

## 4 Lower limit

|            |                                              |
|------------|----------------------------------------------|
| Formula    | $\text{lower} = \bar{x} - ME$                |
| Substitute | $\text{lower} = 100,200 - 5,367.68$          |
| Result     | <b><math>\text{lower} = 94,832.32</math></b> |

## 5 Upper limit

|            |                                               |
|------------|-----------------------------------------------|
| Formula    | $\text{upper} = \bar{x} + ME$                 |
| Substitute | $\text{upper} = 100,200 + 5,367.68$           |
| Result     | <b><math>\text{upper} = 105,567.68</math></b> |

At 95% confidence, the mean data scientist salary lies between \$94,832 and \$105,568.

# The same data at 99% confidence

## 1 New reliability factor

|            |                                                                         |
|------------|-------------------------------------------------------------------------|
| Formula    | $0.99 = 1 - \alpha$ , then look up $z$ at $\alpha/2$                    |
| Substitute | $\alpha = 0.01$ , $\alpha/2 = 0.005$ , table entry $1 - 0.005 = 0.9950$ |
| Look up    | 0.9950 is not listed, so take the nearest: $2.5 + 0.08$                 |
| Result     | <b><math>z = 2.58</math></b>                                            |

## 2 Margin of error

|            |                                   |
|------------|-----------------------------------|
| Formula    | $ME = z \times SE$                |
| Substitute | $ME = 2.58 \times 2,738.61$       |
| Result     | <b><math>ME = 7,065.62</math></b> |

## 3 The two limits

|        |                                                |
|--------|------------------------------------------------|
| Lower  | $100,200 - 7,065.62 = 93,134.38$               |
| Upper  | $100,200 + 7,065.62 = 107,265.62$              |
| Result | <b><math>93,134 \rightarrow 107,266</math></b> |

Width at 95% is \$10,735. Width at 99% is \$14,131. Same sample, more confidence, wider interval.

# A 95% interval from nine salaries

Nine data scientist salaries are collected. The sample mean is \$92,533 and the standard error works out to \$4,644. The population variance is not known. Build a 95% confidence interval.

| GIVEN | n | $\bar{x}$ | SE      | Confidence |
|-------|---|-----------|---------|------------|
|       | 9 | \$92,533  | \$4,644 | 95%        |

1

Recover s

|            |                                                         |
|------------|---------------------------------------------------------|
| Formula    | $SE = s / \sqrt{n}, \text{ so } s = SE \times \sqrt{n}$ |
| Substitute | $s = 4,644 \times \sqrt{9} = 4,644 \times 3$            |
| Result     | <b><math>s = 13,932</math></b>                          |

2

Degrees of freedom and t

|            |                                                            |
|------------|------------------------------------------------------------|
| Formula    | $df = n - 1, \text{ then look up } t \text{ at } \alpha/2$ |
| Substitute | $df = 9 - 1 = 8, \alpha/2 = 0.025$                         |
| Look up    | t-table row $df = 8$ , the 95% confidence interval column  |
| Result     | <b><math>df = 8, t = 2.31</math></b>                       |

## Margin of error and the two limits

### 3 Margin of error

|            |                                    |
|------------|------------------------------------|
| Formula    | $ME = t \times SE$                 |
| Substitute | $ME = 2.31 \times 4,644$           |
| Result     | <b><math>ME = 10,727.64</math></b> |

### 4 Lower limit

|            |                                              |
|------------|----------------------------------------------|
| Formula    | $\text{lower} = \bar{x} - ME$                |
| Substitute | $\text{lower} = 92,533 - 10,727.64$          |
| Result     | <b><math>\text{lower} = 81,805.36</math></b> |

### 5 Upper limit

|            |                                               |
|------------|-----------------------------------------------|
| Formula    | $\text{upper} = \bar{x} + ME$                 |
| Substitute | $\text{upper} = 92,533 + 10,727.64$           |
| Result     | <b><math>\text{upper} = 103,260.64</math></b> |

At 95% confidence the mean lies between \$81,805 and \$103,261. Wider than Problem 1, because the variance had to be estimated.



# Blood magnesium before and after a pill

Ten people have blood magnesium measured before and after taking a pill. The mean of the ten differences is 0.33 mg/dL with a standard deviation of 0.45. Build a 95% interval for the mean difference.

| GIVEN | n pairs | $\bar{d}$ | $s^d$ | Confidence |
|-------|---------|-----------|-------|------------|
|       | 10      | 0.33      | 0.45  | 95%        |

|   |                   |         |                                                                |
|---|-------------------|---------|----------------------------------------------------------------|
| 1 | Collapse the pair | Formula | $d = \text{after} - \text{before}$ , computed for each subject |
|   |                   | Effect  | two columns become one, so a single-sample formula now applies |
|   |                   | Result  | <b>1 variable, n = 10</b>                                      |

|   |                |            |                                                |
|---|----------------|------------|------------------------------------------------|
| 2 | Standard error | Formula    | $SE = s^d / \sqrt{n}$                          |
|   |                | Substitute | $SE = 0.45 / \sqrt{10}$                        |
|   |                | Compute    | $\sqrt{10} = 3.1623$ , so $SE = 0.45 / 3.1623$ |
|   |                | Result     | <b>SE = 0.1423</b>                             |

# Reliability factor through to the interval

## 3 Reliability factor

|            |                                               |
|------------|-----------------------------------------------|
| Formula    | $df = n - 1$ , then look up $t$ at $\alpha/2$ |
| Substitute | $df = 10 - 1 = 9$ , $\alpha/2 = 0.025$        |
| Result     | <b><math>t = 2.26</math></b>                  |

## 4 Margin of error

|            |                                 |
|------------|---------------------------------|
| Formula    | $ME = t \times SE$              |
| Substitute | $ME = 2.26 \times 0.1423$       |
| Result     | <b><math>ME = 0.3216</math></b> |

## 5 The two limits

|        |                                           |
|--------|-------------------------------------------|
| Lower  | $0.33 - 0.3216 = 0.0084$                  |
| Upper  | $0.33 + 0.3216 = 0.6516$                  |
| Result | <b><math>0.01 \rightarrow 0.65</math></b> |

The interval is entirely above zero. With 95% certainty magnesium levels rose, so the pill most likely works.

# Engineering versus management grades

100 engineering students average 58% with a known  $\sigma$  of 10. 70 management students average 65% with a known  $\sigma$  of 5. Build a 95% interval for the difference between the two means.

| GIVEN | $n_1 / n_2$ | $\bar{x} / \bar{y}$ | $\sigma_1 / \sigma_2$ | Confidence |
|-------|-------------|---------------------|-----------------------|------------|
|       | 100 / 70    | 58% / 65%           | 10 / 5                | 95%        |

|   |                |            |                                      |
|---|----------------|------------|--------------------------------------|
| 1 | Point estimate | Formula    | point estimate = $\bar{x} - \bar{y}$ |
|   |                | Substitute | 58 – 65                              |
|   |                | Result     | -7                                   |

|   |                    |             |                                                        |
|---|--------------------|-------------|--------------------------------------------------------|
| 2 | Each variance term | Formula     | $\sigma^2 / n$ , worked out for each sample separately |
|   |                    | Engineering | $10^2 / 100 = 100 / 100 = 1$                           |
|   |                    | Management  | $5^2 / 70 = 25 / 70 = 0.3571$                          |
|   |                    | Result      | 1 and 0.3571                                           |

# Standard error through to the interval

## 3 Standard error

|            |                                                 |
|------------|-------------------------------------------------|
| Formula    | $SE = \sqrt{(\sigma^2_1/n_1 + \sigma^2_2/n_2)}$ |
| Substitute | $SE = \sqrt{(1 + 0.3571)} = \sqrt{1.3571}$      |
| Result     | <b>SE = 1.1650</b>                              |

## 4 Margin of error

|            |                                                             |
|------------|-------------------------------------------------------------|
| Formula    | $ME = z \times SE, \text{ with } z = 1.96 \text{ at } 95\%$ |
| Substitute | $ME = 1.96 \times 1.1650$                                   |
| Result     | <b>ME = 2.2833</b>                                          |

## 5 The two limits

|        |                         |
|--------|-------------------------|
| Lower  | $-7 - 2.2833 = -9.2833$ |
| Upper  | $-7 + 2.2833 = -4.7167$ |
| Result | <b>-9.28 → -4.72</b>    |

The interval is negative because the difference was taken as engineering minus management. Reversing it gives 4.72 to 9.28.

# Apple prices, New York versus LA

10 New York shops average \$3.94 with  $s = 0.18$ . 8 LA shops average \$3.25 with  $s = 0.27$ . Variances are unknown but assumed equal. Build a 95% interval for the difference.

|       | $n_1 / n_2$ | $\bar{x} / \bar{y}$ | $s_1 / s_2$ | Confidence |
|-------|-------------|---------------------|-------------|------------|
| GIVEN | 10 / 8      | \$3.94 / \$3.25     | 0.18 / 0.27 | 95%        |

1

Pooled variance

|            |                                                                  |
|------------|------------------------------------------------------------------|
| Formula    | $s_p^2 = [ (n_1-1)s_1^2 + (n_2-1)s_2^2 ] / ( n_1 + n_2 - 2 )$    |
| Substitute | $s_p^2 = [ 9 \times 0.18^2 + 7 \times 0.27^2 ] / ( 10 + 8 - 2 )$ |
| Compute    | $[ 9(0.0324) + 7(0.0729) ] / 16 = 0.8019 / 16$                   |
| Result     | <b><math>s_p^2 = 0.0501</math></b>                               |

2

Standard error

|            |                                                      |
|------------|------------------------------------------------------|
| Formula    | $SE = \sqrt{ (s_p^2/n_1 + s_p^2/n_2) }$              |
| Substitute | $SE = \sqrt{ (0.00501 + 0.00626) } = \sqrt{0.01127}$ |
| Result     | <b><math>SE = 0.1062</math></b>                      |

# Degrees of freedom through to the interval

## 3 Degrees of freedom and t

|            |                                                             |
|------------|-------------------------------------------------------------|
| Formula    | $df = n_1 + n_2 - 2$ (total size minus number of variables) |
| Substitute | $df = 10 + 8 - 2 = 16, \alpha/2 = 0.025$                    |
| Result     | <b><math>df = 16, t = 2.12</math></b>                       |

## 4 Estimate and margin

|            |                                          |
|------------|------------------------------------------|
| Estimate   | $\bar{x} - \bar{y} = 3.94 - 3.25 = 0.69$ |
| Substitute | $ME = t \times SE = 2.12 \times 0.1062$  |
| Result     | <b><math>ME = 0.2251</math></b>          |

## 5 The two limits

|        |                                           |
|--------|-------------------------------------------|
| Lower  | $0.69 - 0.2251 = 0.4649$                  |
| Upper  | $0.69 + 0.2251 = 0.9151$                  |
| Result | <b><math>0.46 \rightarrow 0.92</math></b> |

**Apples in New York cost between 46 and 92 cents more than in LA, at 95% confidence.**

## SECTION B

# Hypothesis Tests

- One mean, variance known and unknown
- Dependent samples
- Independent samples, both variance cases

# Is the mean salary really \$113,000?

Glassdoor reports a mean data scientist salary of \$113,000. A sample of 30 gives a mean of \$100,200, with a known population  $\sigma$  of \$15,000. Test the claim at 5% significance.

| GIVEN | $H_0$           | $H_1$              | Type      | $\alpha$ |
|-------|-----------------|--------------------|-----------|----------|
|       | $\mu = 113,000$ | $\mu \neq 113,000$ | two-sided | 0.05     |

|   |                |            |                                             |
|---|----------------|------------|---------------------------------------------|
| 1 | Standard error | Formula    | $SE = \sigma / \sqrt{n}$                    |
|   |                | Substitute | $SE = 15,000 / \sqrt{30} = 15,000 / 5.4772$ |
|   |                | Result     | <b>SE = 2,738.61</b>                        |

|   |                 |            |                     |
|---|-----------------|------------|---------------------|
| 2 | Numerator first | Formula    | $\bar{X} - \mu_0$   |
|   |                 | Substitute | $100,200 - 113,000$ |
|   |                 | Result     | <b>-12,800</b>      |



# Comparing against the critical value

## 3 Test statistic

|            |                               |
|------------|-------------------------------|
| Formula    | $Z = (\bar{x} - \mu_0) / SE$  |
| Substitute | $Z = -12,800 / 2,738.61$      |
| Result     | <b><math>Z = -4.67</math></b> |

## 4 Critical value

|         |                                                              |
|---------|--------------------------------------------------------------|
| Rule    | two-sided at $\alpha = 0.05$ , so look up $\alpha/2 = 0.025$ |
| Look up | table entry 0.9750 $\rightarrow$ 1.9 + 0.06                  |
| Result  | <b>critical value = 1.96</b>                                 |

## 5 Compare

|            |                                                                        |
|------------|------------------------------------------------------------------------|
| Rule       | reject $H_0$ when $ Z  >$ critical value                               |
| Substitute | $ -4.67  = 4.67$ versus 1.96                                           |
| Result     | <b><math>4.67 &gt; 1.96 \rightarrow</math> reject <math>H_0</math></b> |

There is no statistical evidence that the mean salary is \$113,000. At 1% the critical value is 2.58, so it is rejected there too.

# Is the competitor's open rate above 40%?

Ten of a competitor's campaign open rates give a sample mean of 37.7% with a standard error of 4.34%. The company's own rate is 40%. Test at 5% significance whether the competitor's rate is higher.

|       |                 |              |           |    |
|-------|-----------------|--------------|-----------|----|
| GIVEN | $H_0$           | $H_1$        | Type      | n  |
|       | $\mu \leq 40\%$ | $\mu > 40\%$ | one-sided | 10 |

|   |                    |            |               |
|---|--------------------|------------|---------------|
| 1 | Degrees of freedom | Formula    | $df = n - 1$  |
|   |                    | Substitute | $df = 10 - 1$ |
|   |                    | Result     | $df = 9$      |

|   |                 |            |                   |
|---|-----------------|------------|-------------------|
| 2 | Numerator first | Formula    | $\bar{X} - \mu_0$ |
|   |                 | Substitute | $37.7 - 40$       |
|   |                 | Result     | $-2.3$            |

# The null stands

## 3 Test statistic

|            |                               |
|------------|-------------------------------|
| Formula    | $T = (\bar{x} - \mu_0) / SE$  |
| Substitute | $T = -2.3 / 4.34$             |
| Result     | <b><math>T = -0.53</math></b> |

## 4 Critical value

|         |                                                                   |
|---------|-------------------------------------------------------------------|
| Rule    | one-sided at $\alpha = 0.05$ , so the whole 0.05 sits in one tail |
| Look up | t-table row $df = 9$ , the one-sided 0.05 column                  |
| Result  | <b>critical value = 1.83</b>                                      |

## 5 Compare

|            |                                                                          |
|------------|--------------------------------------------------------------------------|
| Rule       | reject $H_0$ when $ T  > \text{critical value}$                          |
| Substitute | $ -0.53  = 0.53$ versus 1.83                                             |
| Result     | <b><math>0.53 &lt; 1.83 \rightarrow \text{cannot reject } H_0</math></b> |

**No statistical basis for claiming the competitor's rate is above 40%. The p-value is 0.304, above every common  $\alpha$ .**

# Does the magnesium pill actually work?

The same ten before-and-after readings from Problem 3, now as a test. The difference is taken as before minus after, giving a mean difference of  $-0.33$  with  $s^d = 0.45$ .

|       |            |         |           |           |
|-------|------------|---------|-----------|-----------|
| GIVEN | $H_0$      | $H_1$   | $\bar{d}$ | $n$ pairs |
|       | $D \geq 0$ | $D < 0$ | $-0.33$   | $10$      |

|   |                |         |                                                                                    |
|---|----------------|---------|------------------------------------------------------------------------------------|
| 1 | Restate with D | Start   | $H_0: \mu \text{ before } \geq \mu \text{ after}$                                  |
|   |                | Rewrite | $\mu \text{ before } - \mu \text{ after } \geq 0$ , and that expression is named D |
|   |                | Result  | $H_0: D \geq 0, \quad H_1: D < 0, \quad \text{with } D_0 = 0$                      |

|   |                |            |                                         |
|---|----------------|------------|-----------------------------------------|
| 2 | Standard error | Formula    | $SE = s^d / \sqrt{n}$                   |
|   |                | Substitute | $SE = 0.45 / \sqrt{10} = 0.45 / 3.1623$ |
|   |                | Result     | $SE = 0.1423$                           |

# One statistic, two verdicts

## 3 Test statistic

|            |                               |
|------------|-------------------------------|
| Formula    | $T = (\bar{d} - D_0) / SE$    |
| Substitute | $T = (-0.33 - 0) / 0.1423$    |
| Result     | <b><math>T = -2.32</math></b> |

## 4 Find the p-value

|        |                                                             |
|--------|-------------------------------------------------------------|
| Rule   | one-sided area beyond the statistic, at $df = 9$            |
| Locate | 2.32 sits between the 0.025 and 0.01 columns of the t-table |
| Result | <b>p-value <math>\approx 0.024</math></b>                   |

## 5 Compare at both levels

|                    |                                                      |
|--------------------|------------------------------------------------------|
| At $\alpha = 0.05$ | $0.024 < 0.05 \rightarrow \text{reject } H_0$        |
| At $\alpha = 0.01$ | $0.024 > 0.01 \rightarrow \text{cannot reject } H_0$ |
| Result             | <b>the verdict depends on <math>\alpha</math></b>    |

The smallest significance level at which the null can still be rejected is 0.024, which is exactly the p-value.

# Testing the dean's four-point claim

The dean claims management students outperform engineering by exactly 4 percentage points. Engineering:  $n = 100$ , mean 58%,  $\sigma = 10$ . Management:  $n = 70$ , mean 65%,  $\sigma = 6$ . Test at 5% significance.

|       |                      |                         |           |          |
|-------|----------------------|-------------------------|-----------|----------|
| GIVEN | $H_0$                | $H_1$                   | Type      | $\alpha$ |
|       | $\mu_e - \mu_m = -4$ | $\mu_e - \mu_m \neq -4$ | two-sided | 0.05     |

|   |                |            |                                                 |
|---|----------------|------------|-------------------------------------------------|
| 1 | Standard error | Formula    | $SE = \sqrt{(\sigma^2_1/n_1 + \sigma^2_2/n_2)}$ |
|   |                | Substitute | $SE = \sqrt{(10^2 / 100 + 6^2 / 70)}$           |
|   |                | Compute    | $\sqrt{(1 + 0.5143)} = \sqrt{1.5143}$           |
|   |                | Result     | <b>SE = 1.2306</b>                              |

|   |                 |            |                             |
|---|-----------------|------------|-----------------------------|
| 2 | Numerator first | Formula    | $(\bar{x} - \bar{y}) - D_0$ |
|   |                 | Substitute | $(58 - 65) - (-4) = -7 + 4$ |
|   |                 | Result     | <b>-3</b>                   |

# The sign carries extra information

## 3 Test statistic

Formula  $Z = [ (\bar{x} - \bar{y}) - D_0 ] / SE$

Substitute  $Z = -3 / 1.2306$

Result  **$Z = -2.44$**

## 4 p-value and decision

Rule two-sided, so double the tail area, then compare with  $\alpha$

Substitute  $p\text{-value} = 0.015$  versus  $\alpha = 0.05$

Result  **$0.015 < 0.05 \rightarrow \text{reject } H_0$**

## 5 Read the sign

Rule a negative statistic means the true value is below the hypothesised one

Apply  $Z$  is negative and the hypothesised difference was  $-4$

Result **true difference is likely below  $-4$ , so  $-5$ ,  $-6$  and further**

**Management students outperform engineering by more than 4 percentage points, significant at the 5% level.**

# Testing the apple price difference

The same apple price data from Problem 5, now as a test of whether the two cities charge the same. The pooled work has already been done, so it is reused.

|       |                       |                          |                                                            |    |
|-------|-----------------------|--------------------------|------------------------------------------------------------|----|
| GIVEN | $H_0$                 | $H_1$                    | $D_0$                                                      | df |
|       | $\mu_{ny} = \mu_{la}$ | $\mu_{ny} \neq \mu_{la}$ | 0                                                          | 16 |
| 1     | Carry over the SE     | From Prob. 5             | $s^2_p = 0.0501, \quad SE = \sqrt{(0.0501/10 + 0.0501/8)}$ |    |
|       |                       | Result                   | <b>SE = 0.1062</b>                                         |    |
|       |                       |                          |                                                            |    |
| 2     | Numerator first       | Formula                  | $(\bar{x} - \bar{y}) - D_0$                                |    |
|       |                       | Substitute               | $(3.94 - 3.25) - 0 = 0.69 - 0$                             |    |
|       |                       | Result                   | <b>0.69</b>                                                |    |



# An extreme statistic, and what it means

## 3 Test statistic

Formula  $T = [ (\bar{x} - \bar{y}) - D_0 ] / SE$

Substitute  $T = 0.69 / 0.1062$

Result  **$T = 6.50$**

## 4 Compare

Rule reject  $H_0$  when  $|T| > \text{critical value}$

Substitute 6.50 versus  $t$  at  $df = 16 = 2.12$

Result  **$6.50 \gg 2.12 \rightarrow \text{reject at every level}$**

## 5 Read it carefully

Guide any Z or T score above 4 is extremely significant

Caution means of 3.94 and 3.25 with s around 0.2 already made this obvious

Result **the hypothesis was weak, not brilliantly confirmed**

**A better question would be whether New York prices are 20% higher than LA, rather than simply different.**

## SECTION C

# Practice Problems

- Four problems with raw data and full solutions
- Mean, standard deviation and standard error computed from scratch
- Answer key at the end

## Commute times, variance unknown

A commuter records travel time in minutes on eight randomly chosen weekdays. Times are assumed normally distributed. Build a 95% confidence interval for the true mean commute time.

### DATA

32      28      35      30      27      33      31      30

### What to find

- The sample mean
- The sample variance and standard deviation
- The standard error
- The 95% confidence interval

### Hints

- Population variance is not given, so the statistic is  $t$
- Divide the sum of squared deviations by  $n - 1$ , not by  $n$
- Degrees of freedom are  $8 - 1 = 7$

# Mean, sum of squares and standard deviation

## 1 Sample mean

|            |                                                     |
|------------|-----------------------------------------------------|
| Formula    | $\bar{x} = \Sigma x / n$                            |
| Substitute | $\bar{x} = (32+28+35+30+27+33+31+30) / 8 = 246 / 8$ |
| Result     | <b><math>\bar{x} = 30.75</math></b>                 |

## 2 Sum of squared deviations

|            |                                                                           |
|------------|---------------------------------------------------------------------------|
| Formula    | $\Sigma (x - \bar{x})^2$                                                  |
| Substitute | $(32-30.75)^2 + (28-30.75)^2 + \dots + (30-30.75)^2$                      |
| Compute    | $1.5625 + 7.5625 + 18.0625 + 0.5625 + 14.0625 + 5.0625 + 0.0625 + 0.5625$ |
| Result     | <b><math>\Sigma (x - \bar{x})^2 = 47.50</math></b>                        |

## 3 Variance and std. deviation

|            |                                                                |
|------------|----------------------------------------------------------------|
| Formula    | $s^2 = \Sigma (x - \bar{x})^2 / (n - 1), \quad s = \sqrt{s^2}$ |
| Substitute | $s^2 = 47.50 / (8 - 1) = 47.50 / 7 = 6.7857$                   |
| Result     | <b><math>s = \sqrt{6.7857} = 2.6049</math></b>                 |

# Standard error through to the interval

4

## Standard error

Formula

$$SE = s / \sqrt{n}$$

Substitute

$$SE = 2.6049 / \sqrt{8} = 2.6049 / 2.8284$$

Result

$$SE = 0.9210$$

5

## Margin of error

Look up

$$df = 8 - 1 = 7, \alpha/2 = 0.025 \rightarrow t = 2.365$$

Substitute

$$ME = t \times SE = 2.365 \times 0.9210$$

Result

$$ME = 2.1781$$

6

## The two limits

Lower

$$30.75 - 2.1781 = 28.5719$$

Upper

$$30.75 + 2.1781 = 32.9281$$

Result

$$28.57 \rightarrow 32.93$$

At 95% confidence, the true mean commute time is between 28.57 and 32.93 minutes.

# Bottle filling, variance known

A filling machine is calibrated to 500 ml, and the population standard deviation is known to be 4 ml. A sample of 36 bottles averages 498 ml. Test at 5% significance whether the machine is off target.

|       |                        |                           |     |          |
|-------|------------------------|---------------------------|-----|----------|
| GIVEN | $H_0$                  | $H_1$                     | $n$ | $\sigma$ |
|       | $\mu = 500 \text{ ml}$ | $\mu \neq 500 \text{ ml}$ | 36  | 4 ml     |

## 1 Standard error

|            |                                 |
|------------|---------------------------------|
| Formula    | $SE = \sigma / \sqrt{n}$        |
| Substitute | $SE = 4 / \sqrt{36} = 4 / 6$    |
| Result     | <b><math>SE = 0.6667</math></b> |

## 2 Test statistic

|            |                                                     |
|------------|-----------------------------------------------------|
| Formula    | $Z = (\bar{x} - \mu_0) / SE$                        |
| Substitute | numerator = $498 - 500 = -2$ , so $Z = -2 / 0.6667$ |
| Result     | <b><math>Z = -3.00</math></b>                       |

# Decision, then the same answer from an interval

## 3 Compare

|            |                                                                   |
|------------|-------------------------------------------------------------------|
| Rule       | reject $H_0$ when $ Z  > \text{critical value}$                   |
| Substitute | $ -3.00  = 3.00$ versus 1.96                                      |
| Result     | <b><math>3.00 &gt; 1.96 \rightarrow \text{reject } H_0</math></b> |

## 4 Build the interval instead

|            |                                               |
|------------|-----------------------------------------------|
| Formula    | $CI = \bar{x} \pm z \times SE$                |
| Substitute | $498 \pm 1.96 \times 0.6667 = 498 \pm 1.3067$ |
| Result     | <b><math>496.69 \rightarrow 499.31</math></b> |

## 5 Cross-check

|        |                                                                        |
|--------|------------------------------------------------------------------------|
| Rule   | if the hypothesised value falls outside the interval, the test rejects |
| Apply  | 500 ml is not between 496.69 and 499.31                                |
| Result | <b>both methods reject <math>H_0</math></b>                            |

**The machine is underfilling. A two-sided test at  $\alpha$  and an interval at  $1 - \alpha$  always give the same verdict.**

## Paired scores before and after training

Six employees take an assessment before and after a training programme. Build a 95% confidence interval for the mean improvement, then test whether the programme works.

| Employee           | 1  | 2  | 3  | 4  | 5  | 6  |
|--------------------|----|----|----|----|----|----|
| Before             | 68 | 72 | 75 | 70 | 66 | 74 |
| After              | 74 | 75 | 79 | 73 | 71 | 78 |
| d = after – before | 6  | 3  | 4  | 3  | 5  | 4  |

### Why this is a one-sample problem

The differences in the bottom row are what get analysed. Two columns collapse into one, so every formula from the single-mean case applies without change.



# Mean difference and standard deviation

## 1 Mean difference

|            |                                                  |
|------------|--------------------------------------------------|
| Formula    | $\bar{d} = \Sigma d / n$                         |
| Substitute | $\bar{d} = (6 + 3 + 4 + 3 + 5 + 4) / 6 = 25 / 6$ |
| Result     | <b><math>\bar{d} = 4.1667</math></b>             |

## 2 Sum of squared deviations

|            |                                                            |
|------------|------------------------------------------------------------|
| Formula    | $\Sigma (d - \bar{d})^2$                                   |
| Substitute | $(6 - 4.1667)^2 + (3 - 4.1667)^2 + \dots + (4 - 4.1667)^2$ |
| Compute    | $3.3611 + 1.3611 + 0.0278 + 1.3611 + 0.6944 + 0.0278$      |
| Result     | <b><math>\Sigma (d - \bar{d})^2 = 6.8333</math></b>        |

## 3 Variance and std. deviation

|            |                                                                        |
|------------|------------------------------------------------------------------------|
| Formula    | $s^{2d} = \Sigma (d - \bar{d})^2 / (n - 1), \quad s^d = \sqrt{s^{2d}}$ |
| Substitute | $s^{2d} = 6.8333 / 5 = 1.3667$                                         |
| Result     | <b><math>s^d = \sqrt{1.3667} = 1.1690</math></b>                       |

# The interval, then the test

## 4 Standard error

|            |                                            |
|------------|--------------------------------------------|
| Formula    | $SE = s^d / \sqrt{n}$                      |
| Substitute | $SE = 1.1690 / \sqrt{6} = 1.1690 / 2.4495$ |
| Result     | <b>SE = 0.4773</b>                         |

## 5 Confidence interval

|            |                                                                      |
|------------|----------------------------------------------------------------------|
| Look up    | $df = 6 - 1 = 5, \alpha/2 = 0.025 \rightarrow t = 2.571$             |
| Substitute | $ME = 2.571 \times 0.4773 = 1.2270, \text{ then } 4.1667 \pm 1.2270$ |
| Result     | <b>2.94 → 5.39</b>                                                   |

## 6 Test statistic

|            |                                |
|------------|--------------------------------|
| Formula    | $T = (\bar{d} - D_0) / SE$     |
| Substitute | $T = (4.1667 - 0) / 0.4773$    |
| Result     | <b>T = 8.73 &gt;&gt; 2.571</b> |

The interval sits entirely above zero and T far exceeds the critical value. The training programme works.

# Two branches, unknown but equal variances

Daily sales at Branch A: n = 6, mean 42, s = 5. Branch B: n = 7, mean 37, s = 4. Variances are unknown but assumed equal. Build a 95% interval for the difference and test it at 5%.

| GIVEN | $n_1 / n_2$ | $\bar{x} / \bar{y}$ | $s_1 / s_2$ | $\alpha$ |
|-------|-------------|---------------------|-------------|----------|
|       | 6 / 7       | 42 / 37             | 5 / 4       | 0.05     |

1

Pooled variance

|            |                                                               |
|------------|---------------------------------------------------------------|
| Formula    | $s_p^2 = [ (n_1-1)s_1^2 + (n_2-1)s_2^2 ] / ( n_1 + n_2 - 2 )$ |
| Substitute | $s_p^2 = [ 5 \times 5^2 + 6 \times 4^2 ] / ( 6 + 7 - 2 )$     |
| Compute    | $( 125 + 96 ) / 11 = 221 / 11$                                |
| Result     | <b><math>s_p^2 = 20.0909</math></b>                           |

2

Standard error

|            |                                                     |
|------------|-----------------------------------------------------|
| Formula    | $SE = \sqrt{ ( s_p^2/n_1 + s_p^2/n_2 ) }$           |
| Substitute | $SE = \sqrt{ ( 3.3485 + 2.8701 ) } = \sqrt{6.2186}$ |
| Result     | <b><math>SE = 2.4937</math></b>                     |

# The interval, then the test

## 3 Confidence interval

|            |                                                                         |
|------------|-------------------------------------------------------------------------|
| Look up    | $df = 6 + 7 - 2 = 11, \alpha/2 = 0.025 \rightarrow t = 2.201$           |
| Substitute | estimate = 5, $ME = 2.201 \times 2.4937 = 5.4887$ , then $5 \pm 5.4887$ |
| Result     | <b>-0.49 → 10.49</b>                                                    |

## 4 Test statistic

|            |                                        |
|------------|----------------------------------------|
| Formula    | $T = [(\bar{x} - \bar{y}) - D_0] / SE$ |
| Substitute | $T = (5 - 0) / 2.4937$                 |
| Result     | <b>T = 2.01</b>                        |

## 5 Compare

|            |                                                         |
|------------|---------------------------------------------------------|
| Rule       | reject $H_0$ when $ T  > \text{critical value}$         |
| Substitute | 2.01 versus 2.201                                       |
| Result     | <b>2.01 &lt; 2.201 → cannot reject <math>H_0</math></b> |

Zero sits inside the interval and T falls short of the critical value. No significant difference between the branches.

## SECTION D

# Your Turn: Activity

- Eight problems covering every case in this unit
- No worked solutions — only final answers at the very end
- Show the formula, the substitution and the result for each step

# How to work and what to hand in

Every problem is solved the same way you have watched me solve fourteen of them. Show the same three lines for every step.

1

## Write the formula

In symbols first, before any numbers appear. This is worth marks on its own, even if the arithmetic later goes wrong.

2

## Substitute

Rewrite the same formula with the actual values in place. Show the intermediate arithmetic, such as  $\sqrt{30} = 5.4772$ .

3

## State the result

Keep four decimals through the working and round only at the end. Carry the result into the next step.

4

## Answer in a sentence

Never stop at two numbers. Say what the interval means, or whether the null is rejected and at what level.

### Reminders

- Unless a problem says otherwise, use 95% confidence and a 5% significance level
- Every critical value you need is on the reference slide near the end of this deck
- For the tests, state the null and the alternative before computing anything

# Problems 1 and 2

1

## Courier delivery times

A courier company has tracked delivery times for years and knows the population standard deviation is 6 minutes. A fresh sample of 25 deliveries averages 34 minutes. Delivery times are normally distributed.

 $n = 25$  $\bar{x} = 34 \text{ min}$  $\sigma = 6 \text{ min}$ 

FIND

The 95% confidence interval for the true mean delivery time.

2

## Sari-sari store sales

Daily sales at a small store, recorded in hundreds of pesos over seven days. The population variance is not known, and sales are assumed normally distributed.

 $n = 7$ 

12 15 11 14 13 16 12

FIND

The mean,  $s$ , the standard error, and the 95% confidence interval.

*Problem 2 gives you raw numbers, so the mean and the standard deviation have to be built from scratch before anything else.*

## Problem 3

Five students had their typing speed measured in words per minute before and after a short course. Build a 95% confidence interval for the mean improvement, then test at 5% significance whether the course actually works.

| Student            | 1  | 2  | 3  | 4  | 5  |
|--------------------|----|----|----|----|----|
| Before             | 32 | 28 | 35 | 30 | 27 |
| After              | 38 | 33 | 39 | 36 | 31 |
| d = after – before | ?  | ?  | ?  | ?  | ?  |

### Part A · the interval

- Fill in the difference row first
- Find  $\bar{d}$ ,  $s^d$  and the standard error
- Build the 95% interval and check whether zero is inside it

### Part B · the test

- State  $H_0$  and  $H_1$  for a one-sided test
- Compute T using the same standard error
- Compare against the critical value and decide



## Problems 4 and 5

4

### Waiting times, two branches

A bank compares customer waiting times at two branches. Both population standard deviations are known from past records. The samples are independent.

**Branch X:**  $n = 50$ ,  $\bar{x} = 8.4$ ,  $\sigma = 2$

**Branch Y:**  $n = 40$ ,  $\bar{y} = 7.1$ ,  $\sigma = 1.5$

FIND

**The 95% confidence interval for the difference in mean waiting time.**

5

### Two exam sections

Two sections sat the same exam. Population variances are unknown, but since both sections had the same teacher and the same paper, assume they are equal.

**Section A:**  $n = 8$ ,  $\bar{x} = 78$ ,  $s = 6$

**Section B:**  $n = 10$ ,  $\bar{y} = 72$ ,  $s = 5$

FIND

**The pooled variance, the 95% interval, and a 5% test of whether the sections differ.**

*Problem 5 needs the pooled variance before anything else, and remember the degrees of freedom are  $n_1 + n_2 - 2$ .*

## Problems 6 and 7

6

### Cereal box weight

A machine is set to fill boxes to 350 grams, and the population standard deviation is known to be 8 grams. A quality check on 40 boxes gives a sample mean of 347 grams. Test whether the machine is off target.

$$n = 40$$

$$\bar{x} = 347 \text{ g}$$

$$\sigma = 8 \text{ g}$$

FIND

$H_0$  and  $H_1$ , the Z-score, and the decision at 5% significance.

7

### Tutorial centre claim

A tutorial centre advertises that its students average above 80 in a national exam. A sample of 12 of its students gives a mean of 82.5 with a sample standard deviation of 5.2. Test the claim.

$$n = 12$$

$$\bar{x} = 82.5$$

$$s = 5.2$$

FIND

$H_0$  and  $H_1$ , the T-score, and the decision at 5% significance.

*Problem 6 is two-sided because either direction would be a fault. Problem 7 is one-sided, and the null must state the opposite of the advertised claim.*

## Problem 8

8

### The branch manager's claim

Using the same bank waiting time data from Problem 4, a manager claims that Branch X takes exactly one minute longer than Branch Y on average. Test that specific claim at 5% significance.

Branch X:  $n = 50$ ,  $\bar{x} = 8.4$ ,  $\sigma = 2$

Branch Y:  $n = 40$ ,  $\bar{y} = 7.1$ ,  $\sigma = 1.5$

FIND

$H_0$  and  $H_1$ , the Z-score, and the decision.

Read this before you start

**The hypothesised difference here is NOT zero.**

- The manager named a specific number, so  $D_0$  is 1, not 0
- That value has to be subtracted in the numerator of the statistic
- The standard error is identical to Problem 4, so carry it over rather than recomputing
- Decide which mean you subtract from which, then stay consistent

When you have finished all eight, check your final answers against the key on the last slide. If a number does not match, the fault is almost always in the standard error, so start there.

# Critical values used in this deck

## z • standard normal

|                            |              |
|----------------------------|--------------|
| one-sided, $\alpha = 0.05$ | <b>1.645</b> |
| two-sided, $\alpha = 0.05$ | <b>1.96</b>  |
| one-sided, $\alpha = 0.01$ | <b>2.33</b>  |
| two-sided, $\alpha = 0.01$ | <b>2.58</b>  |

## t • two-sided, $\alpha = 0.05$

|        |              |
|--------|--------------|
| df = 5 | <b>2.571</b> |
| df = 7 | <b>2.365</b> |
| df = 8 | <b>2.306</b> |
| df = 9 | <b>2.262</b> |

## More t values used in these problems

|              |              |              |                   |
|--------------|--------------|--------------|-------------------|
| df = 11      | df = 16      | df = 22      | df = 9, one-sided |
| <b>2.201</b> | <b>2.120</b> | <b>2.074</b> | <b>1.833</b>      |

Answer keys sometimes round these, for example 2.31 in place of 2.306. Either value gives the same conclusion.

# All ten worked problems at a glance

|    |                                 |                                                    |                      |
|----|---------------------------------|----------------------------------------------------|----------------------|
| 1  | CI, $\sigma$ known              | SE 2,738.61 · z 1.96 · ME 5,367.68                 | 94,832 → 105,568     |
| 2  | CI, $\sigma$ unknown            | SE 4,644 · t 2.31, df 8 · ME 10,727.64             | 81,805 → 103,261     |
| 3  | CI, dependent                   | SE 0.1423 · t 2.26, df 9 · ME 0.3216               | 0.01 → 0.65          |
| 4  | CI, independent, $\sigma$ known | SE 1.1650 · z 1.96 · ME 2.2833                     | -9.28 → -4.72        |
| 5  | CI, independent, pooled         | $s^2_{\bar{X}}$ 0.0501 · SE 0.1062 · t 2.12, df 16 | 0.46 → 0.92          |
| 6  | Z-test, $\sigma$ known          | Z -4.67 versus 1.96                                | reject $H_0$         |
| 7  | t-test, $\sigma$ unknown        | T -0.53 versus 1.83, p 0.304                       | cannot reject        |
| 8  | t-test, dependent               | T -2.32, p 0.024                                   | reject at 5%, not 1% |
| 9  | Z-test, independent             | Z -2.44, p 0.015                                   | reject $H_0$         |
| 10 | t-test, pooled                  | T 6.50 versus 2.12                                 | reject at all levels |

Practice problems: 1) 28.57 → 32.93 2) Z -3.00, reject 3) 2.94 → 5.39, T 8.73, reject 4) -0.49 → 10.49, T 2.01, cannot reject

# Six mistakes that cost the most marks

1

## Dividing by $n$ instead of $\sqrt{n}$

The standard error divides the standard deviation by the square root of the sample size.

2

## Using $n$ instead of $n - 1$

Sample variance divides the sum of squares by  $n - 1$ . Only population variance divides by  $n$ .

3

## Forgetting to halve $\alpha$

A two-sided test at 5% looks up 0.025, not 0.05.

4

## Dropping the hypothesised value

When  $D_0$  is not zero, it must be subtracted in the numerator of the statistic.

5

## Wrong degrees of freedom

One sample gives  $n - 1$ . Two independent samples give  $n_1 + n_2 - 2$ .

6

## Comparing raw signs

Compare absolute values against the critical value, then read the sign for direction.

# Final answers only — remove this slide before handing out

|   |                 |                                                     |                              |
|---|-----------------|-----------------------------------------------------|------------------------------|
| 1 | Courier times   | SE 1.2000 · z 1.96 · ME 2.3520                      | 31.65 → 36.35 min            |
| 2 | Sari-sari sales | $\bar{x}$ 13.2857 · s 1.7995 · SE 0.6801 · t 2.447  | 11.62 → 14.95                |
| 3 | Typing speed    | d 6,5,4,6,4 · $\bar{d}$ 5 · $s^d$ 1.0 · SE 0.4472   | 3.76 → 6.24, T 11.18, reject |
| 4 | Bank branches   | SE 0.3691 · z 1.96 · ME 0.7235                      | 0.58 → 2.02 min              |
| 5 | Exam sections   | $s^2_{\bar{x}}$ 29.8125 · SE 2.5899 · t 2.12, df 16 | 0.51 → 11.49, T 2.32, reject |
| 6 | Cereal boxes    | SE 1.2649 · Z -2.37 versus 1.96                     | reject $H_0$                 |
| 7 | Tutorial centre | SE 1.5011 · T 1.67 versus 1.796, df 11              | cannot reject $H_0$          |
| 8 | Manager's claim | $D_0 = 1$ · Z 0.81 versus 1.96                      | cannot reject $H_0$          |

Problem 8 is the one to check carefully. Forgetting to subtract  $D_0 = 1$  gives  $Z = 3.52$  and a wrong rejection.